

COSC 594/650

Math review

Dongsheng Ding

dongshed@utk.edu

Department of Electrical Engineering and Computer Science
University of Tennessee, Knoxville

Terminology

Decision variable: $x \in \mathbb{R}^n$

Objective function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$

Feasible region: $X \subseteq \mathbb{R}^n$

■ OPTIMIZATION PROBLEM

minimize $f(x)$
 $x \in \mathbb{R}^n$
subject to $x \in X$

GOAL: Find a **best** decision

Outline

- LINEAR ALGEBRA
- PROBABILITY
- MATHEMATICAL ANALYSIS

Linear algebra

Notation

■ SET OPERATION

- ★ $X_1 \cap X_2$ – intersection
- ★ $X_1 \cup X_2$ – union
- ★ $X_1 \setminus X_2$ – difference
- ★ $X_1 \times X_2$ – Cartesian product

■ INFIMUM AND SUPREMUM

- ★ $\sup X$ – least upper bound
- ★ $\inf X$ – greatest lower bound

■ FUNCTION

- ★ $f: X \rightarrow Y$ – mapping from domain X to range Y

Examples of supremum and infimum

■ SUPREMUM

$$\star S_1 = \{1, 2, 5, 9\}$$

$$\sup S_1 = \max S_1$$

$$\star S_2 = \{x \in \mathbb{R} \mid x^2 < 2\}$$

$$\sup S_2 = \sqrt{2} \notin S_2$$

$$\star S_3 = \{\sin(n) \mid n \in \mathbb{N}\}$$

$$\sup S_3 = 1 \notin S_3$$

■ INFIMUM

$$\star T_1 = \{-3, 1, 4\}$$

$$\inf T_1 = \min T_1$$

$$\star T_2 = \{\tfrac{1}{n} \mid n \in \mathbb{N}\}$$

$$\inf T_2 = 0 \notin T_2$$

$$\star T_3 = \{\sin(n) \mid n \in \mathbb{N}\}$$

$$\inf T_3 = -1 \notin T_3$$

Vector, subspace, and norm

■ LINEAR COMBINATION

- ★ $ax + by$, $x, y \in \mathbb{R}^n$, $a, b \in \mathbb{R}$ – linear combination

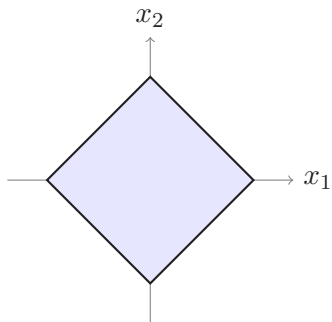
■ SUBSPACE AND LINEAR INDEPENDENCE

- ★ $V \subset \mathbb{R}^n$ – subspace of $\mathbb{R}^n$ (non-empty, and closure under addition and scalar multiplication)
- ★ $a_1v_1 + \dots + a_mv_m = 0$, $v_i \in V$, $a_i \in \mathbb{R} \implies a_i = 0$ – linear independence

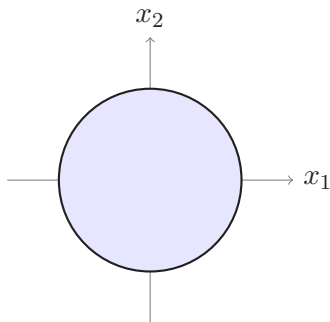
■ VECTOR NORM

- ★ $x^\top y = \sum_{i=1}^n x_i y_i$ or $\langle x, y \rangle$, $x, y \in \mathbb{R}^n$ – inner product
- ★ $x^\top y = 0$ or $x \perp y = 0$, $x, y \in \mathbb{R}^n$ – orthogonality
- ★ $\sqrt{x^\top x} = \sqrt{\sum_{i=1}^n x_i^2}$ or $\|x\|_2$ – ℓ_2 -norm

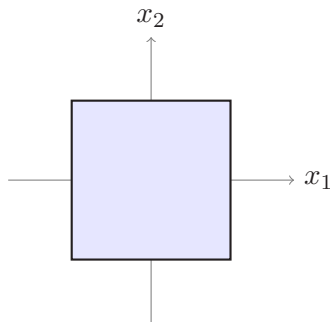
ℓ_1 , ℓ_2 , and ℓ_∞ norms



$$\|x\|_1 = |x_1| + |x_2|$$



$$\|x\|_2 = \sqrt{x_1^2 + x_2^2}$$



$$\|x\|_\infty = \max\{|x_1|, |x_2|\}$$

■ NORM INEQUALITY

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$$

Vector inequalities

■ CAUCHY-SCHWARZ INEQUALITY

$$x^\top y \leq \|x\| \|y\|$$

★ $\cos \theta = \frac{x^\top y}{\|x\| \|y\|}$ – angle between two vectors (orthogonal, parallel, antiparallel)

★ $x^\top y = \|x\| \|y\|$ – equality

■ TRIANGLE INEQUALITY

$$\|x + y\| \leq \|x\| + \|y\|$$

★ $\|x + y\| = \|x\| + \|y\|$ – equality

Matrix

■ MATRIX OPERATIONS

- ★ $C = A + B$, $A, B \in \mathbb{R}^{m \times n}$, i.e., $C_{ij} = A_{ij} + B_{ij}$ – matrix addition
- ★ $C = AB$, $A \in \mathbb{R}^{m \times p}$, $B \in \mathbb{R}^{p \times n}$, i.e., $C_{ij} = \sum_{k=1}^p A_{ik} B_{kj}$ – matrix product
- ★ $A \in \mathbb{R}^{m \times m}$ – square matrix
- ★ $\langle A, B \rangle = \sum_{i,j} A_{ij} B_{ij}$ or $\text{tr}(A^\top B)$ – matrix inner product

■ EIGENVECTOR AND EIGENVALUE

$$Av = \lambda v, \quad A \in \mathbb{R}^{m \times m} (\text{square matrix}), v \in \mathbb{R}^m, \lambda \in \mathbb{R}$$

- ★ $v \in \mathbb{R}^m$, $\lambda \in \mathbb{R}$ – eigenvector and eigenvalue
- ★ $\rho(A) = \max_i \{|\lambda_i| : Av_i = \lambda_i v_i\}$ – spectral radius
- ★ $A = \sum_{i=1}^m \lambda_i v_i (v_i)^\top$ or $V \Lambda V^\top$, $A = A^\top \in \mathbb{R}^m$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_m)$ – eigendecomposition

Matrix

■ SINGULAR VALUE

$$A^\top A v_i = \sigma_i^2 v_i \quad \text{or} \quad A A^\top u_i = \sigma_i^2 u_i, \quad \sigma_i \geq 0$$

- ★ $\sigma_1 \geq \dots \geq \sigma_p \geq 0$, $A \in \mathbb{R}^{m \times n}$, $p = \min\{m, n\}$ – singular values
- ★ $A = \sum_{i=1}^p \sigma_i u_i v_i^\top = U \Sigma V^\top$, $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_p)$, $U^\top U = V^\top V = I_p$ – singular value decomposition

■ MATRIX NORM

- ★ $\|A\|_2 = \sup_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \sigma_1$ or $\|A\|$ – induced ℓ_2 -norm
- ★ $\|A\|_F = \sqrt{\sum_{i,j} |A_{ij}|^2} = \sqrt{\sum_{i=1}^p \sigma_i^2}$ – Frobenius norm
- ★ $\|A\|_* = \sum_{i=1}^p \sigma_i$ – nuclear norm

Matrix inequalities

■ NORM HIERARCHY

$$\|A\|_* \geq \|A\|_F \geq \|A\|_2$$

$$\|A\|_2 \geq \rho(A), \quad \text{square matrix } A$$

■ INDUCED NORM INEQUALITY

$$\|Ax\| \leq \|A\| \|x\|$$

Matrix inequalities

■ SUBMULTIPLICATIVITY

$$\|AB\| \leq \|A\|\|B\|$$

■ QUADRATIC-FORM BOUNDS

$$\lambda_{\min}(A)\|x\|^2 \leq x^\top Ax \leq \lambda_{\max}(A)\|x\|^2, \quad \text{symmetric matrix } A = A^\top$$

- ★ $\lambda_{\min}(A) \geq 0$ – positive semi-definite (PSD) matrix
- ★ $\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$, $\sigma_i(A) = |\lambda_i(A)|$ – condition number

Mathematical analysis

Big O notation

Function to be estimated: $f(x)$

Comparison function: $g(x)$

Asymptotic direction: $x \rightarrow \infty$

■ ASYMPTOTICS

- ★ $f(x) = \mathcal{O}(g(x))$, i.e., $\exists \alpha, x_0, |f(x)| \leq \alpha |g(x)|, \forall x > x_0$ – big O
- ★ $f(x) = \Omega(g(x))$, i.e., $\exists \alpha, x_0, |f(x)| \geq \alpha |g(x)|, \forall x > x_0$ – big omega
- ★ $f(x) = o(g(x))$, i.e., $\forall \alpha > 0, \exists x_0, |f(x)| \leq \alpha |g(x)|, \forall x > x_0$ – small o
- ★ $f(x) = \omega(g(x))$, i.e., $\forall \alpha > 0, \exists x_0, |f(x)| \geq \alpha |g(x)|, \forall x > x_0$ – small omega

Derivative

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (differentiable)

Derivative:
$$\frac{\partial f(x)}{\partial x_i} = \lim_{t \rightarrow 0} \frac{f(x + te_i) - f(x)}{t}$$

■ GRADIENTS

★ $\nabla f(x) = \left[\frac{\partial f(x)}{\partial x_1} \quad \frac{\partial f(x)}{\partial x_2} \quad \cdots \quad \frac{\partial f(x)}{\partial x_n} \right]^\top$ – derivative

★ $\nabla^2 f(x) = \left[\frac{\partial^2 f(x)}{\partial x_i \partial x_j} \right]$ – Hessian

★ Taylor expansion

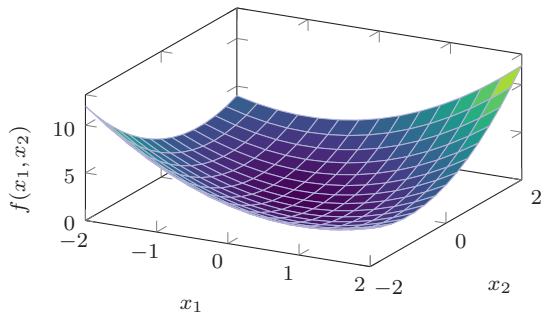
$$f(x) = f(x_0) + \nabla f(x_0)^\top (x - x_0) + \frac{1}{2} (x - x_0)^\top \nabla^2 f(x_0) (x - x_0) + o(\|x - x_0\|^2)$$

Example

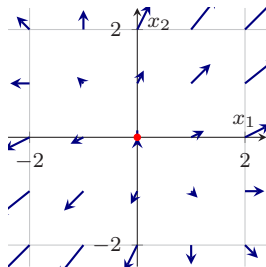
■ QUADRATIC FUNCTION

$$f(x_1, x_2) = x_1^2 + x_1x_2 + x_2^2 \quad \nabla f(x) = \begin{bmatrix} 2x_1 + x_2 \\ x_1 + 2x_2 \end{bmatrix} \quad \nabla^2 f(x) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Function surface: $f(x_1, x_2)$



Gradient field: $\nabla f(x)$



Mean value theorem

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (differentiable)

■ REPRESENTATIONS

★ linear function

$$f(x) = f(x_0) + \nabla f(\xi)^\top (x - x_0), \quad \xi \in [x_0, x]$$

★ quadratic function

$$f(x) = f(x_0) + \nabla f(x_0)^\top (x - x_0) + \frac{1}{2} (x - x_0)^\top \nabla^2 f(\xi) (x - x_0), \quad \xi \in [x_0, x]$$

Chain rule

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ (differentiable)

Jacobian: $\nabla f(x) = [\nabla f_1(x) \nabla f_2(x) \dots \nabla f_m(x)]$

■ CHAIN RULE

★ $g(x): \mathbb{R}^m \rightarrow \mathbb{R}^p, h(x) = g(f(x))$

$$\nabla h(x) = \nabla f(x) \nabla g(f(x))$$

★ $f(x) = Ax$

$$\nabla h(x) = A^\top \nabla g(Ax)$$

$$\nabla^2 h(x) = A^\top \nabla^2 g(Ax) A$$

Contraction mapping

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ (differentiable)

Lipschitz continuity: $\|f(x) - f(y)\| \leq L\|x - y\|, \forall x, y$

■ CONTRACTION

★ $f(x): \mathbb{R}^n \rightarrow \mathbb{R}^n, L < 1$ – contractive mapping

$$x \rightarrow f(x) \rightarrow f(f(x)) \rightarrow \dots \rightarrow x^*$$

★ $x^* = f(x^*)$ – existence and uniqueness of a fixed point

Probability

Probability notion

Probability: $P(X = x) \geq 0, \sum_x P(X = x) = 1$

Random variable: $P(X \in E)$

■ PROBABILITY RULES

- ★ $\mathbb{E}[X] = \sum_x xP(X = x)$ – expectation
- ★ $\text{Var}(X) = \mathbb{E}[(x - \mathbb{E}[X])^2]$ – variance
- ★ $P(X = x, Y = y) = P(X = x)P(Y = y)$ – independent random variables
- ★ $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])^\top]$ – covariance matrix

Similar concepts in continuous space

Gaussian

Gaussian: $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, or $X \sim \mathcal{N}(\mu, \sigma^2)$

Joint Gaussian: $f(x) = \frac{1}{\sqrt{(2\pi)^n \det(\Sigma)}} e^{-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)}$, or $X \sim \mathcal{N}(\mu, \Sigma)$

■ PROBABILITY RULES

- ★ $\mathbb{E}[X] = \mu$ – expectation
- ★ $\text{Cov}(X, X) = \Sigma$ – covariance matrix
- ★ joint Gaussian: $\text{Cov}(X, Y) = 0 \Leftrightarrow$ independence

Convergence

Random sampling: $X_1, X_2, X_3, \dots$ i.i.d. ($\mathbb{E}[X_i] = \mu$, $\text{Cov}(X_i, X_j) = \Sigma$)

Sample mean: $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$

■ LAW OF LARGE NUMBERS

★ $\bar{X}_n \xrightarrow{P} \mu, n \rightarrow \infty$ – weak law

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| < \epsilon) = 1$$

★ $\bar{X}_n \xrightarrow{a.s.} \mu, n \rightarrow \infty$ – strong law

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1$$

Convergence

Random sampling: $X_1, X_2, X_3, \dots$ i.i.d. ($\mathbb{E}[X_i] = \mu$, $\text{Cov}(X_i, X_j) = \Sigma$)

Sample mean: $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$

■ CENTRAL LIMIT THEOREM

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{D} \mathcal{N}(0, \Sigma)$$

Probability inequalities

■ MARKOV'S INEQUALITY

★ RV $X \geq 0$

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

■ CHEBYSHEV'S INEQUALITY

★ RV X , $\mathbb{E}[X] = \mu$, $\text{Var}(X) = \sigma^2$

$$P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}$$

■ CAUCHY-SCHWARZ INEQUALITY

$$|\mathbb{E}[XY]|^2 \leq \mathbb{E}[X^2] \mathbb{E}[Y^2]$$

Questions?

(**Email: dongshed@utk.edu**)